

GPS, an NVIDIA Jetson Nano board, cameras and sensors. There are plans to launch it by the end of this year. He has also started the *IoETPlanet.com*, a collaborative platform to bring IoT enthusiasts together for the first time under a single umbrella where they can learn, collaborate and contribute.

## Large e-governance projects and startups in India run on open source

The open source space has really evolved, as we all know. Raina says, “Gone are the days when we had to campaign for the community to use open source. Today, we are in a new era where open source has become the new normal. With the democratisation of technologies like cloud computing, containers, serverless and artificial intelligence, open source has become much more mainstream. Today, some of the largest e-governance projects and startups in India are running on open source.”

Raina adds, “If you look at Stack Overflow’s recent survey results, India represents the second largest developer base on this site. As per the report, developers in India are the most likely to say their organisations are using blockchain technology. I am confident that within this decade India will become the largest base of developers. Let’s not forget that India now ranks among the Top 10 countries in terms of the number of open source projects.”

But there are ways in which the country can improve still further. Raina insists that the AICTE (All India Council for Technical Education) should introduce new courses on some of the emerging technologies like Docker, Kubernetes,



serverless and IoT to upskill students and get them ready before their job searches. Second, it should encourage students to use Slack or IRC to spend more time with open source developers while they are studying for their engineering degrees.

## Apart from certification courses and exams, contributions to OSS also count

Raina adds, “It is really important for companies in India to identify what open source software they are using and whether they are in compliance with all the licences. If MNCs buy open source software to run their desktops in-house, and they do not sell that product, there is probably no danger of violating the terms of the OSS’ licence. Often, however, companies are at risk if they incorporate open source software tools into a product they end up selling.”

Raina has a smart tip for all engineering students. They should not limit themselves to just taking certification courses and exams, but also contribute towards open source software. “We are still lagging behind in terms

of the number of open source projects; hence, we need to get more involved, contribute, collaborate and convert ideas into URLs,” Raina says.

Regarding the future of open source, Raina says, “If you ask me to name one open source tool that every cloud vendor unanimously relies on today, it is Kubernetes. It is making multi-cloud adoption painless. With the emergence of the Cloud Native Computing Foundation (CNCF), Docker and Kubernetes, the future of OSS looks promising. CNCF is part of the non-profit Linux Foundation and serves as the vendor-neutral home for many of the fastest growing open source projects, including Kubernetes, Prometheus, and Envoy. It brings together the world’s top developers, end users, and vendors and runs the largest open source developer conferences. Undoubtedly, open source software is a key enabler in all the areas of technology that we encounter now and in the future.” 

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